

Отчет по лабораторной работе №2 по ком. сетям

1. Установили ansible на ноут и создали пользователя на VPS с sudo правами ansible

```
namil27@namil27-MaiBook-X-series:~$ ansible --version
ansible 2.10.8
config file = None
```

2. Создали inventory.ini

```
GNU nano 0.2
[vds]
server ansible_host=178.72.131.136 ansible_user=ansible
```

3. Балуемся с командами как в практике:

```
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m ping
server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m file -a "path=/tmp/ansible_test.txt state=touch"
server | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "dest": "/tmp/ansible_test.txt",
  "gid": 1005,
  "group": "ansible",
  "mode": "0664",
  "owner": "ansible",
  "size": 0,
  "state": "file",
  "uid": 1005
}
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m shell -a "ls -l /tmp/ansible_test.txt"
server | CHANGED | rc=0 >>
-rw-rw-r-- 1 ansible ansible 0 Dec 23 15:18 /tmp/ansible_test.txt
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m copy -a "content='Hello from Ansible ad-hoc' dest=/tmp/ansible_test.txt"
server | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "checksum": "5dffb08066ff2678c4d7ab13095566f4f4576a595",
  "dest": "/tmp/ansible_test.txt",
  "gid": 1005,
  "group": "ansible",
  "md5sum": "08ec1e05e4a2d5349d8b15be51ca7cba",
  "mode": "0664",
  "owner": "ansible",
  "size": 25,
  "src": "/home/ansible/.ansible/tmp/ansible-tmp-1766503170.7469826-94297-110376899738608/source",
  "state": "file",
  "uid": 1005
}
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m shell -a "cat /tmp/ansible_test.txt"
server | CHANGED | rc=0 >>
Hello from Ansible ad-hoc
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m file -a "path=/tmp/ansible_test.txt state=absent"
server | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "path": "/tmp/ansible_test.txt",
  "state": "absent"
}
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m shell -a "ls /tmp/ansible_test.txt"
server | FAILED | rc=1 >>
ls: cannot access '/tmp/ansible_test.txt': No such file or directoryman: no return code
```

4. Теперь балуемся с плейбуком как в примере лабы (цикл жизни файла)

```
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ nano file_playbook.yml
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible-playbook -i inventory.ini file_playbook.yml

PLAY [File operations playbook] *****

TASK [Gathering Facts] *****
ok: [server]

TASK [Create file] *****
changed: [server]

TASK [Write initial content] *****
changed: [server]

TASK [Modify file content before deletion] *****
changed: [server]

TASK [Delete file] *****
changed: [server]

PLAY RECAP *****
server                : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

5. устанавливаем caddy и запускаем

```
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible-playbook -i inventory.ini caddy_install.yml
PLAY [Install and start Caddy] *****
TASK [Gathering Facts] *****
ok: [server]
TASK [Install dependencies] *****
changed: [server]
TASK [Add Caddy GPG Key] *****
[WARNING]: Consider using the get_url or uri module rather than running 'curl'. If you need to use command because get_url or uri is insufficient you can
add 'warn: false' to this command task or set 'command_warnings=False' in ansible.cfg to get rid of this message.
changed: [server]
TASK [Add Caddy repository] *****
changed: [server]
TASK [Update apt cache] *****
changed: [server]
TASK [Install Caddy] *****
changed: [server]
TASK [Ensure Caddy is running] *****
ok: [server]
PLAY RECAP *****
server:          : ok=7    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m shell -a "systemctl status caddy --no-pager"
server | CHANGED | rc=0 >>
● caddy.service - Caddy
   Loaded: loaded (/lib/systemd/system/caddy.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-23 15:29:38 UTC; 18s ago
     Docs: https://caddyserver.com/docs/
    Main PID: 1757359 (caddy)
      Tasks: 7 (limit: 2262)
    Memory: 11.7M
       CPU: 99ms
    CGroup: /system.slice/caddy.service
            └─1757359 /usr/bin/caddy run --environ --config /etc/caddy/Caddyfile

namil27@namil27-MaiBook-X-series:~/Учеба/compnet$
```

← ↻ 178.72.131.136 Caddy works! 📖 🌐 Перевести ⌵ 🇷🇺 🇺🇸 🇯🇵 🇮🇹 🇪🇸 🇫🇷 🇩🇪 🇬🇧 🇮🇳 🇮🇩 🇻🇳 🇰🇷 🇹🇼 🇦🇺 🇨🇦 🇬🇧 🇮🇳 🇮🇩 🇻🇳 🇰🇷 🇹🇼 🇦🇺 🇨🇦



Congratulations! おめでとう! ¡Felicidades! 恭喜! बधाई हो!
Поздравляю! Вітаю! Gefeliciteerd! Glückwunsch! 🎉

Your web server is working. Now make it work for you. 🍌

Caddy is ready to serve your site over HTTPS:

1. Point your domain's A/AAAA DNS records at this machine.
2. Upload your site's files to `/var/www/html`.
3. Edit your Caddyfile at `/etc/caddy/Caddyfile`:
 - a. Replace `:80` with your domain name
 - b. Change the site root to `/var/www/html`
4. Reload the configuration: `systemctl reload caddy`
5. Visit your site!

If that worked 🥳

Awesome! You won't have to look at this slanted page anymore.

Caddy can do a lot more than serve static files. It's also a

6. Зарегистрируем домен на duckdns и настроим https, а также поменяем index.html на свой

```
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m file -a "path=/var/www/html state=directory mode=0755" -b
server | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/var/www/html",
  "size": 4096,
  "state": "directory",
  "uid": 0
}
```

```
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ nano index.html
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m copy -a "src=index.html dest=/var/www/html/index.html mode=0644" -b
server | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "checksum": "46af1845185c7f7a77df335572bf45c13b2e81d4",
  "dest": "/var/www/html/index.html",
  "gid": 0,
  "group": "root",
  "md5sum": "d9b7b19fb8d49472692441b78127b40",
  "mode": "0644",
  "owner": "root",
  "size": 215,
  "src": "/home/ansible/.ansible/tmp/ansible-tmp-1766504662.0146024-95896-182025601298856/source",
  "state": "file",
  "uid": 0
}
```

смотрим caddy файл

```
namil27@namil27-MaiBook-X-series:~/Учеба/compnet$ ansible -i inventory.ini vds -m shell -a "cat /etc/caddy/Caddyfile" -b
server | CHANGED | rc=0 >>
# The Caddyfile is an easy way to configure your Caddy web server.
#
# Unless the file starts with a global options block, the first
# uncommented line is always the address of your site.
#
# To use your own domain name (with automatic HTTPS), first make
# sure your domain's A/AAAA DNS records are properly pointed to
# this machine's public IP, then replace ":80" below with your
# domain name.

:80 {
    # Set this path to your site's directory.
    root * /usr/share/caddy

    # Enable the static file server.
    file_server

    # Another common task is to set up a reverse proxy:
    # reverse_proxy localhost:8080

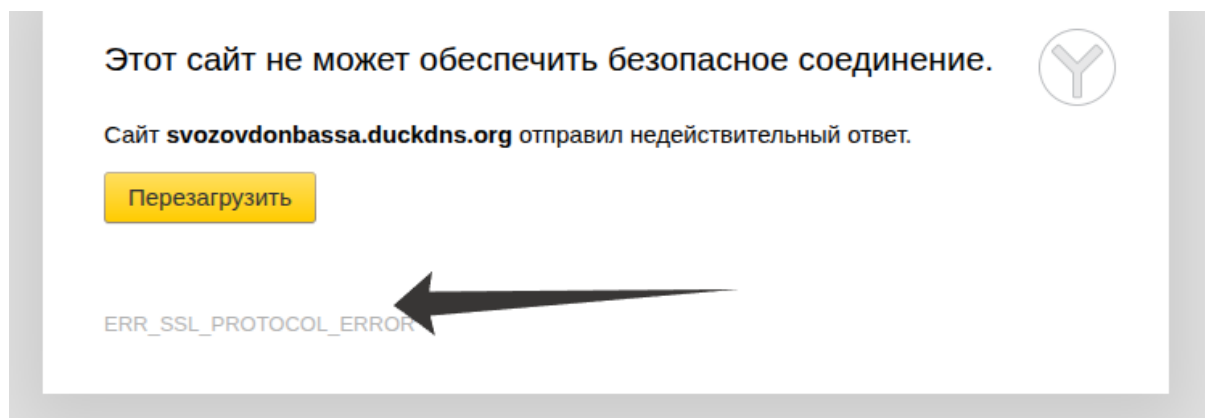
    # Or serve a PHP site through php-fpm:
    # php_fastcgi localhost:9000
}

# Refer to the Caddy docs for more information:
# https://caddyserver.com/docs/caddyfile
```

надо поменять чтобы заработал https, caddy сам все намотит

```
GNU nano 6.2
svozovdonbassa.duckdns.org {
    root * /var/www/html
    file_server
}
```

создали файл на ноутбуке и щас просто с копируем его на сервер через ansible и рестартнем caddy и все должно заработать Но нет!



Ищем проблему и находим в логах caddy, и...

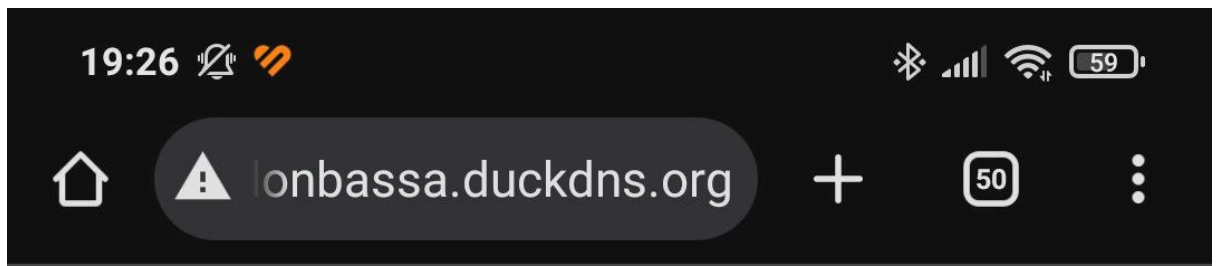
DNS problem: SERVFAIL looking up CAA for duckdns.org

Резюмирую:

В процессе настройки HTTPS с использованием Let's Encrypt была выявлена проблема со стороны DNS-провайдера DuckDNS — серверы сертификации возвращали ошибку SERVFAIL при проверке CAA-записей домена. В связи с этим получение сертификата оказалось невозможным.

Получается с бесплатным доменом не вышло получить сертификат, поэтому просто поменяем index.html.

Так как Яндекс браузер принципиально не открывает стараничку по https я зашел по адресу <http://svozovdonbassa.duckdns.org> с браузера google с телефона и прикладываю скриншот



Hello from Ansible

HTTPS works!

Domain: svovzdombassa.duckdns.org

7. Сделаем задание 1 (добавили все в один файл, изменили содержимое файла и добавили редактировать перед удалением)

```
GNU nano 6.2
- name: File operations playbook (task 1)
  hosts: vds
  become: true

  vars:
    target_file: /tmp/ansible_playbook.txt

  tasks:
    - name: Create file
      file:
        path: "{{ target_file }}"
        state: touch

    - name: Check that file exists
      stat:
        path: "{{ target_file }}"
        register: file_stat

    - name: Fail if file was not created
      fail:
        msg: "File was not created: {{ target_file }}"
        when: not file_stat.stat.exists

    - name: Write initial content
      copy:
        dest: "{{ target_file }}"
        content: "Initial content from Ansible playbook\n"

    - name: Modify file content before deletion
      copy:
        dest: "{{ target_file }}"
        content: "Modified content before deletion\n"

    - name: Show final content (proof for report)
      command: cat "{{ target_file }}"
      register: cat_out
      changed_when: false

    - name: Print final content
      debug:
        msg: "{{ cat_out.stdout }}"

    - name: Delete file
      file:
        path: "{{ target_file }}"
        state: absent
```

8. Задание №2 не делаем, так как мы делали работу сразу на 2х машинах